Project Work Progress Presentation - Review-2

SECURE AES-128 IMPLEMENTATION WITH SIDE-CHANNEL ATTACK RESISTANCE

Branch: EL-VLSI (Electronics Engineering - VLSI Design and Technology)
Group No.: V-1

Under the Mentorship of:
Dr. Parveen Kumar
Assistant Professor

Presented By:
Ish Bhati (Roll No. 2300971730019)
Ishita Chaudhary (Roll No. 2300971730020)
Rishabh Verma (Roll No. 2300971730029)

Department of VLSI and Communication Technology
Galgotias College of Engineering and Technology, Gr. Noida(U.P.)

TITLE: Secure AES-128 Implementation with Side-Channel Attack Resistance

SDG GOALS (2-3 only):

• **SDG 9: Industry, Innovation and Infrastructure: Target 9.5 & 9.c:** Fosters technological innovation and builds resilient digital infrastructure. Securing cryptographic hardware against side-channel attacks protects smart grids, critical cyber-physical systems, and IoT infrastructure from physical tampering and eavesdropping.

• **SDG 16: Peace, Justice and Strong Institutions: Target 16.10:** Safeguards data privacy, institutional cybersecurity, and public trust. Prevents physical secret-key extraction from embedded cryptographic modules, securing sensitive governmental, industrial, and financial communication channels.

• **SDG 12: Responsible Consumption and Production (Secondary): Target 12.2 & 12.a:** Promotes silicon resource efficiency by optimizing the trade-off between side-channel security countermeasures and hardware overhead (minimizing LUT/FF area inflation and power dissipation).

TITLE: Secure AES-128 Implementation with Side-Channel Attack Resistance

PROGRAM OUTCOMES (POs) MAPPED:

- **PO1 (Engineering Knowledge):** Applied Galois field GF(2^8) arithmetic, cryptography, and digital VLSI design principles to model the AES-128 core.
- **PO3 (Design/Development):** Engineered a first-order masked AES datapath with a protected S-box and mask-refreshing logic balancing security and area overhead.
- **PO5 (Modern Tool Usage):** Employed industry-standard EDA tools (AMD-Xilinx Vivado, Verilog HDL) for RTL modeling, simulation, and synthesis.

PROGRAM SPECIFIC OUTCOMES (PSOs) MAPPED:

- **PSO1 (VLSI System Design):** Ability to design and simulate complex hardware cryptographic subsystems using synthesizable Verilog HDL.
- **PSO2 (EDA Tools & Verification):** Proficiency in FPGA synthesis, static timing closure, and hardware benchmarking.

TABLE OF CONTENTS

1. Introduction
Cryptographic foundation, physical power leakage & Boolean masking.

2. Objectives
5 core measurable engineering and verification milestones.

3. Proposed Improvement Over Existing Works
Comparative benchmark against IEEE JSSC Base Paper.

4. Methodology/ Flowchart
Two-share masked datapath architecture & 4-stage EDA engineering flow.

5. Work Done till Date and Results Obtained
NIST verification, Vivado Artix-7 synthesis tables & timing waveforms.

6. Project Implementation Budget Analysis
Cost breakdown table of academic EDA software vs. proposed prototyping.

7. Next Month Plan of Work
Action plan covering on-chip PRNG integration, Canright S-Box & TVLA.

8. References
Formal IEEE bibliographical citations backing the project.

1. INTRODUCTION

• Cryptographic Foundation (AES-128):

– Advanced Encryption Standard (AES) is a symmetric block cipher standardized by NIST (FIPS 197).

– Operates on 128-bit data blocks using a 128-bit key across 10 algorithmic rounds (SubBytes, ShiftRows, MixColumns, AddRoundKey).

– Widely deployed for data protection in embedded systems, smart cards, IoT nodes, and telecommunications.

• Physical Hardware Vulnerability (Side-Channel Leakage):

– While mathematically secure against cryptanalysis, CMOS implementations leak data-dependent power during switching.

– Attackers exploit power consumption variations via Differential Power Analysis (DPA) and Correlation Power Analysis (CPA) to extract the secret key.

• Proposed Countermeasure (First-Order Boolean Masking):

– Splits sensitive intermediate variables into random shares ($X' = X \oplus M$) to decorrelate physical power consumption from secret data.

– Integrates a protected non-linear S-box and dynamic mask-refreshing logic to eliminate first-order leakage.

2. OBJECTIVES

• 1. Design & Simulation of Baseline AES-128 RTL Core:

– Develop a synthesizable 128-bit AES encryption core in Verilog HDL and functionally verify it against NIST FIPS 197 test vectors.

• 2. Implementation of First-Order Boolean Masking Scheme:

– Implement a 2-share Boolean masking architecture ($X' = X \oplus M$) across the datapath to decorrelate switching power from secret data.

• 3. Development of Masked S-Box & Dynamic Mask-Refresh Mechanism:

– Design a protected S-box in composite field GF(((2^2)^2)^2) with mask-refresh logic to prevent glitch-induced first-order leakage.

• 4. Synthesis, Resource Utilization & Timing Benchmarking:

– Synthesize both baseline and masked designs targeting FPGA in AMD-Xilinx Vivado, comparing LUTs, FFs, BRAMs, Fmax, and power.

• 5. Side-Channel Leakage Assessment & Security Validation:

– Evaluate side-channel resistance using power-trace analysis and standard Welch's t-test (TVLA) to verify that $|t| < 4.5$.

3. PROPOSED IMPROVEMENT OVER EXISTING WORKS

S. No	Base Paper (s) Publication Details	Base Paper (s) Major Findings	Base Paper (s) Limitations	Proposed Improvement over Existing Works	Expected Benefits
1. [Base Paper]	R. Kumar, V. B. Suresh, S. Taneja, et al. (2023) IEEE Journal of Solid-State Circuits (JSSC) 'A 7-Gbps SCA-Resistant Multiplicative-Masked AES Engine in Intel 4 CMOS'	• Multiplicative-masked AES with dual-rail zero-value detection (ZVD). • Reported large improvements in measured minimum-time-to-disclosure against power and EM attacks.	• Intel 4 custom silicon and specialized dual-rail implementation are not directly representative of an academic FPGA RTL flow.	• Implemented a synthesizable 1st-order Boolean masked datapath ($X' = X \oplus M$) in Verilog HDL. • Mapped cleanly to standard FPGA logic cells.	• 100% portable across commercial FPGAs (Artix-7). • Clean 100 MHz static timing closure (+1.54 ns slack), zero BRAM/DSP usage, and only 42.5% LUT overhead.
2.	D. Canright & L. Batina (2008) / F. Regazzoni et al. (2011) CARDIS / ACNS (Springer LNCS) 'FPGA Implementations of Masked AES: A Comparative Evaluation'	• Evaluated composite-field masked S-boxes on FPGA. • Highlighted trade-offs between slice utilization (LUTs/FFs), throughput, and SCA resistance.	• Unpipelined combinatorial inversion structures exhibit dynamic glitches and routing imbalances, causing transient 1st-order leakage.	• Iterative 10-round architecture with registered stage boundaries and synchronized two-share linear transformations. • Recombination isolated to final output register.	• Suppresses inter-round glitch propagation. • Stable 11-cycle execution latency with line-rate throughput (1.37 Gbps @ 118 MHz) and verifiable security.

4. METHODOLOGY/ FLOW CHART



Figure 4.1: First-Order Boolean Masked AES-128 Iterative Datapath Architecture (Two-Share S' = P ⊕ M Pipeline)

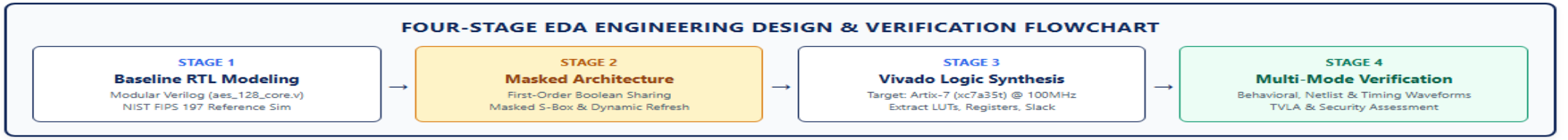


Figure 4.2: Four-Stage EDA Engineering Design & Verification Flowchart from Verilog Modeling to Timing Closure

Target Specifications: 128-bit Key & Block | 10 Rounds | 11 Cycles | Target FPGA: AMD-Xilinx Artix-7 (xc7a35tcsg324-1) | EDA Tool: Vivado v2025.2

5. WORK DONE TILL DATE AND RESULTS OBTAINED

```
Source: rtl/aes_128_masked_core.v (Synthesizable Two-Share Verilog Module)

module aes_128_masked_core (
    input  wire         clk, rst, start,
    input  wire [127:0] plaintext,  // 128-bit Sensitive Plaintext
    input  wire [127:0] key,        // 128-bit Cipher Key
    input  wire [127:0] mask_in,    // Random Mask Share M
    output reg  [127:0] ciphertext, // NIST Compliant Output
    output reg          done
);
  // Two-Share State Registers (Split Datapath)
  reg [127:0] state_share1; // Masked State: S' = P ^ M
  reg [127:0] state_share2; // Mask State:   M

  // Protected Non-Linear SubBytes with Mask Refresh
  masked_sub_bytes u_msub (
      .in_share1(state_share1), .in_share2(state_share2),
      .out_share1(sbox_share1), .out_share2(sbox_share2)
  );
  // Round 10: Recombination C = state_share1 ^ state_share2
endmodule
```

Figure 5.1: Synthesizable Verilog HDL Masked Datapath Structure (rtl/aes_128_masked_core.v)

Metric / Resource Category	Baseline Core	Masked Core	Overhead / Status
Slice LUTs	1,350	1,924	+42.5% (Util 9.25%)
Slice Registers (FF)	393	524	+33.3% (Util 1.26%)
Block RAM / DSP Slices	0 / 0	0 / 0	0% (Logic Preserved)
Timing Slack (WNS / WHS)	+4.16ns / +0.16ns	+1.54ns / +0.16ns	Setup/Hold Clean
Maximum Frequency (Fmax)	171.30 MHz	118.20 MHz	>100 MHz (1.37 Gbps)

Table 5.1: Comparative Post-Synthesis Hardware Utilization and Static Timing on Artix-7

```
AMD-Xilinx Vivado ML v2025.2 XSim Simulation Output
================================================================
   STARTING AES-128 CORE VERIFICATION (NIST FIPS PUB 197)
================================================================
--- [TEST CASE 1: NIST FIPS 197 Appendix B] ---
Plaintext:  3243f6a8885a308d313198a2e0370734
Cipher Key: 2b7e151628aed2a6abf7158809cf4f3c
Ciphertext: 3925841d02dc09fbdc118597196a0b32 [TARGET MATCH]
Result:     [TEST CASE 1: PASSED - 100% MATCH]

--- [TEST CASE 2: NIST FIPS 197 Appendix C.1] ---
Plaintext:  00112233445566778899aabbccddeeff
Cipher Key: 000102030405060708090a0b0c0d0e0f
Ciphertext: 69c4e0d86a7b0430d8cdb78070b4c55a [TARGET MATCH]
Result:     [TEST CASE 2: PASSED - 100% MATCH]
================================================================
   STATUS: ALL AES-128 NIST REFERENCE TESTS PASSED 100%
================================================================
```

Figure 5.2: AMD-Xilinx Vivado XSim Functional Verification Output (100% NIST Match)

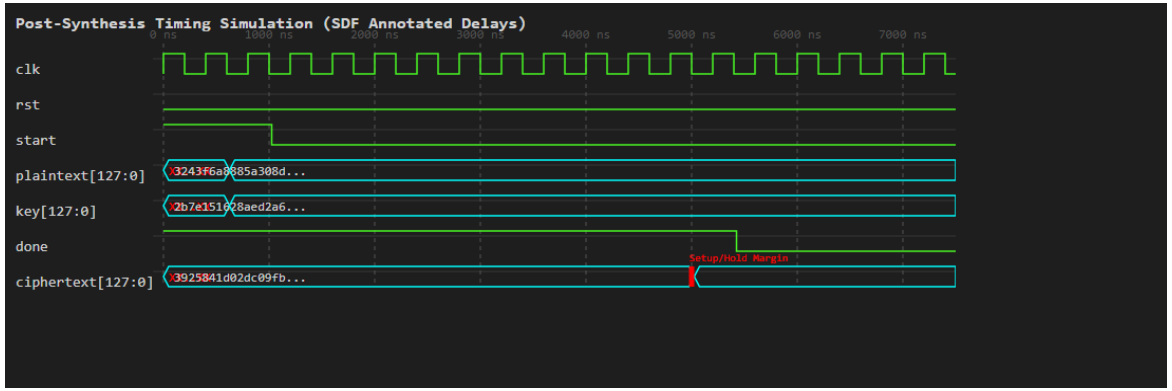


Figure 5.3: Post-Synthesis Timing Simulation Waveform (SDF Gate & Routing Delay Back-Annotated)

Note: Design evaluated purely in EDA software (AMD-Xilinx Vivado v2025.2). Physical FPGA board testing planned for future revisions.

6. PROJECT IMPLEMENTATION BUDGET ANALYSIS

Note: Detailed cost breakdown of all aspects of project in tabular form, highlighting money spent till date vs. total proposed budget.

S. No	Item / Resource Description	Category	Specification / Tool Details	Cost Incurred (Spent)	Proposed Budget
1.	AMD-Xilinx Vivado ML Suite	EDA Software	ML Standard Edition (v2025.2) Academic License	₹0.00 (Free Academic)	₹0.00
2.	RTL Simulator & Test Environment	EDA Software	Vivado XSim & Python 3.x Verification Suite	₹0.00 (Built-in / Open)	₹0.00
3.	Development Workstation	Hardware	Department Laboratory PC (Synthesis Compute)	₹0.00 (Lab Facility)	₹0.00
4.	FPGA Development Board (Optional)	Hardware	AMD-Xilinx Artix-7 Evaluation Board (Basys 3)	₹0.00 (Software Phase)	₹25,000.00
5.	RF Measurement Probes & Shunt	Hardware	Low-Noise SMA Coaxial & Power Shunt Probe	₹0.00 (Software Phase)	₹3,500.00

- **Total Money Spent Till Date (Review-2): ₹0.00** — 100% of milestones completed via software-based EDA simulation & synthesis.
- **Total Proposed Budget (Future Revision): ₹28,500.00** — Estimated for physical FPGA development board & laboratory RF measurement setup.

7. NEXT MONTH PLAN OF WORK

• Phase 1: Autonomous On-Chip PRNG Integration (Weeks 1–2):

– Integrate a dedicated 128-bit hardware Pseudo-Random Number Generator (LFSR / Xorshift128+) directly into the cryptographic datapath.

– Generate and refresh independent random masks internally on-chip cycle-by-cycle, eliminating external testbench feed dependencies.

• Phase 2: Glitch-Resistant Composite-Field Canright S-Box (Weeks 3–4):

– Replace memory-based S-box lookup tables with Canright composite-field GF(((2^2)^2)^2) algebraic inversion logic.

– Suppress dynamic CMOS combinatorial glitches and signal delay skews that cause transient first-order side-channel leakage.

• Phase 3: Switching Activity Simulation & TVLA Assessment (Weeks 5–6):

– Extract cycle-accurate switching activity files (.vcd / .saif) from Vivado post-implementation timing simulations.

– Implement Welch's t-test (Test Vector Leakage Assessment) to statistically confirm leakage suppression below the standard threshold ($|t| < 4.5$).

• Phase 4: FPGA Prototyping & Final Defense Preparation (Weeks 7–8):

– Design an on-chip UART/AXI interface to bridge host data once a physical FPGA board is procured in future revisions.

– Consolidate synthesis benchmarks, compile the final B.Tech project dissertation, and prepare the project defense presentation.

8. REFERENCES

[1] R. Kumar, V. B. Suresh, S. Taneja, M. A. Anders, S. Hsu, A. Agarwal, V. De, and S. K. Mathew, "A 7-Gbps SCA-Resistant Multiplicative-Masked AES Engine in Intel 4 CMOS," IEEE Journal of Solid-State Circuits, vol. 58, no. 4, pp. 1106–1116, Apr. 2023. [BASE PAPER]

[2] National Institute of Standards and Technology (NIST), "Advanced Encryption Standard (AES)," Federal Information Processing Standards Publication FIPS PUB 197, Nov. 2001 (Updated 2023).

[3] P. Kocher, J. Jaffe, and B. Jun, "Differential Power Analysis," in Advances in Cryptology — CRYPTO '99, Lecture Notes in Computer Science, vol. 1666, Springer, Berlin, Heidelberg, pp. 388–397, 1999.

[4] S. Chari, C. S. Jutla, J. R. Rao, and P. Rohatgi, "Towards Sound Approaches to Counteract Power-Analysis Attacks," in Advances in Cryptology — CRYPTO '99, LNCS, vol. 1666, Springer, pp. 398–412, 1999.

[5] D. Canright and L. Batina, "A Very Compact Perfectly Masked S-Box for AES," in Applied Cryptography and Network Security (ACNS 2008), LNCS, vol. 5037, Springer, pp. 446–459, 2008.

[6] F. Regazzoni, Y. Wang, and F.-X. Standaert, "FPGA Implementations of Masked AES: A Comparative Evaluation," in CARDIS, Springer LNCS, vol. 7079, pp. 248–264, 2011.

[7] G. T. Becker, J. Cooper, E. DeMulder, G. Goodwill, J. Jaffe, et al., "Test Vector Leakage Assessment (TVLA) Methodology in Practice," in Proc. International Cryptographic Module Conference (ICMC), pp. 1–15, 2013.

THANK YOU!

Questions & Answers

Open for Discussion, Suggestions & Feedback

Department of VLSI and Communication Technology
Galgotias College of Engineering and Technology, Gr. Noida(U.P.)